

Aliasghar Bazrafkan

GIS, Remote Sensing, and Machine Learning Specialist

Profile

Innovative researcher and GIS analyst with over 10 years of industrial and academic experience in remote sensing, GIS, and Python scripting. Expertise in geoprocessing, spatial analysis, and automated workflows. Skilled in developing geospatial tools and methodologies in ArcGIS Pro for hydrology, precision agriculture, and hyperspectral imagery. Proven ability to create scalable solutions for environmental challenges and enhance decision-making through data-driven insights. Published author and experienced presenter with a strong commitment to innovation and collaboration.

Details

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Key Skills

Technical Expertise

- Geoprocessing: Tool Development, Data Packaging, Python Scripting
- GIS: ArcGIS Pro, ArcGIS Online, QGIS
- Remote Sensing: Satellite & UAS, Image Analysis
- Machine Learning: Random Forest, XGBoost, ANN
- Climate Data Integration: Google Earth Engine
- Aerial Imagery Processing: LiDAR and Hyperspectral Data Processing

Software & Tools

- Programming: Python, R, Java script, HTML
- GIS Tools: ArcGIS, QGIS
- Remote Sensing Tools: ENVI, GDAL,
- Machine Learning Frameworks: TensorFlow, Keras, Pytorch
- Hardware: Drone Piloting (Licensed FAA Part 107)

Employment History

Remote Sensing Researcher

North Dakota State University
June 2022 – Present

- Developed Python-based geoprocessing tools in ArcGIS Pro for hydrological modeling and precision agriculture, improving data analysis efficiency.
- Integrated UAS imagery with satellite data to enhance plant breeding programs, reducing processing time by 50%
- Conducted geoprocessing and feature extraction from hyperspectral imagery to support agricultural and environmental management.
- Led the development of automated decision-making for big datasets, cutting decision time in half.
- Reduced weed detection processing time to 10 minutes for a 16-acre corn field.

GIS Specialist - Internship

North Dakota State University— Statewide Irrigation Feasibility Study
May 2024 – Present

- Created soil irrigability maps using geoprocessing tools, identifying over 17 million acres for potential irrigation projects.
- Automated data packaging workflows in ArcGIS Pro, increasing the accuracy of land suitability assessments.

GIS and Remote Sensing Specialist

*Fars Province Agricultural Organization
January 2014 – June 2022*

- Developed an agricultural atlas using Landsat and Sentinel-2 imagery, improving crop estimation accuracy by 30%.
- Designed soil salinity maps using Landsat and Sentinel-2 imagery with 89% accuracy.
- Conducted satellite image analysis for crop yield prediction and erosion monitoring.

Watershed Manager

*Fars Province Natural Resources Organization
January 2012 – December 2014*

- Designed and implemented spatial workflows for hydrological modeling, achieving an 89% accuracy in forest risk zoning.
- Automated data extraction and conversion workflows for watershed management projects, enhancing project delivery efficiency.

Education

Ph.D. in Natural Resources Sciences

North Dakota State University
June 2022 – April 2025

M.S. in Natural Resources Engineering

Yazd University
January 2010 – January 2012

B.Sc. in Natural Resources Management

Gonbadkaboos University
January 2008 – January 2010

A.S. in Natural Resources Management

Ferdowsi University of Mashhad
January 2006 – January 2008

Professional Certifications

- Licensed Drone Pilot (FAA Certified)
- Advanced Python Programming for Geospatial Applications

Portfolio Link

My Work Portfolio

Professional Profiles



GitHub



LinkedIn



Google Scholar